

```
from google.colab import files
uploaded = files.upload()
```

Choose Files

Online Sales Data.csv

- **Online Sales Data.csv**(text/csv) - 21744 bytes, last modified: 9/30/2025 - 100% done

Saving Online Sales Data.csv to Online Sales Data.csv

```
import sqlite3
import pandas as pd
df = pd.read_csv("Online Sales Data.csv") # Preview data
print("Dataset Preview:")
display(df.head())
```

Dataset Preview:

	Transaction ID	Date	Product Category	Product Name	Units Sold	Unit Price	Total Revenue	Region	Payment Method
0	10001	2024-01-01	Electronics	iPhone 14 Pro	2	999.99	1999.98	North America	Credit Card
1	10002	2024-01-02	Home Appliances	Dyson V11 Vacuum	1	499.99	499.99	Europe	PayPal
2	10003	2024-01-03	Clothing	Levi's 501 Jeans	3	69.99	209.97	Asia	Debit Card
3	10004	2024-01-04	Books	The Da Vinci Code	4	15.99	63.96	North America	Credit Card
4	10005	2024-01-05	Beauty Products	Neutrogena Skincare Set	1	89.99	89.99	Europe	PayPal

```
df.columns

Index(['Transaction ID', 'Date', 'Product Category', 'Product Name',
      'Units Sold', 'Unit Price', 'Total Revenue', 'Region',
      'Payment Method'],
      dtype='object')
```

```
conn = sqlite3.connect('Online Sales.db')
df.to_sql('online_sales', conn, if_exists='replace', index=False)
```

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```
query1 = '''
SELECT
STRFTIME('%Y-%m', "Date") AS YearMonth,
SUM("Total Revenue") AS MonthlyRevenue,
COUNT(DISTINCT "Transaction ID") AS OrderVolume
FROM online_sales
GROUP BY YearMonth
ORDER BY YearMonth
'''

monthly_stats = pd.read_sql_query(query1, conn)
print("Monthly Revenue and Order Volume:")
display(monthly_stats)
```

Monthly Revenue and Order Volume:

	YearMonth	MonthlyRevenue	OrderVolume
0	2024-01	14548.32	31
1	2024-02	10803.37	29
2	2024-03	12849.24	31
3	2024-04	12451.69	30
4	2024-05	8455.49	31
5	2024-06	7384.55	30
6	2024-07	6797.08	31
7	2024-08	7278.11	27

Next steps: [Generate code with monthly_stats](#) [New interactive sheet](#)

```
query2 = '''
SELECT
STRFTIME('%Y-%m', "Date") AS YearMonth,
SUM("Total Revenue") AS MonthlyRevenue,
COUNT(DISTINCT "Transaction ID") AS OrderVolume
FROM online_sales
GROUP BY YearMonth
ORDER BY MonthlyRevenue DESC
LIMIT 5
'''

top_revenue_months = pd.read_sql_query(query3, conn)
print("Top 5 Months by Revenue:")
display(top_revenue_months)
```

Top 5 Months by Revenue:

	YearMonth	MonthlyRevenue	OrderVolume	
0	2024-01	14548.32	31	
1	2024-03	12849.24	31	
2	2024-04	12451.69	30	
3	2024-02	10803.37	29	
4	2024-05	8455.49	31	

Next steps: [Generate code with top_revenue_months](#) [New interactive sheet](#)

```
query3 = '''
SELECT
STRFTIME('%Y-%m', "Date") AS YearMonth,
SUM("Total Revenue") AS MonthlyRevenue,
COUNT(DISTINCT "Transaction ID") AS OrderVolume
FROM online_sales
GROUP BY YearMonth
ORDER BY OrderVolume DESC
LIMIT 5
'''

top_order_months = pd.read_sql_query(query4, conn)
print("Top 5 Months by Order Volume:")
display(top_order_months)
```

Top 5 Months by Order Volume:

	YearMonth	MonthlyRevenue	OrderVolume	
0	2024-07	6797.08	31	
1	2024-05	8455.49	31	
2	2024-03	12849.24	31	
3	2024-01	14548.32	31	
4	2024-06	7384.55	30	

Next steps: [Generate code with top_order_months](#) [New interactive sheet](#)

```
from google.colab import files
import sqlite3
import pandas as pd

uploaded = files.upload()
df = pd.read_csv("Online Sales Data.csv")

# Convert Date to datetime, letting pandas infer format
df["Date"] = pd.to_datetime(df["Date"], errors='coerce')

# Drop rows where Date conversion failed
df = df.dropna(subset=["Date"])

# Format Date as YYYY-MM-DD for SQLite
df["Date"] = df["Date"].dt.strftime('%Y-%m-%d')
```

```
conn = sqlite3.connect(':memory:')
df.to_sql('online_sales', conn, index=False)

monthly_stats = pd.read_sql_query('''
SELECT
STRFTIME('%Y-%m', "Date") AS YearMonth,
SUM("Total Revenue") AS MonthlyRevenue,
COUNT(DISTINCT "Transaction ID") AS OrderVolume
FROM online_sales
GROUP BY YearMonth
ORDER BY YearMonth
''', conn)

display(monthly_stats)

top_revenue = monthly_stats.sort_values(by='MonthlyRevenue', ascending=False).
display(top_revenue)

top_orders = monthly_stats.sort_values(by='OrderVolume', ascending=False).head
display(top_orders)

conn.close()
```

Choose Files Online Sales Data.csv

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Saving Online Sales Data.csv to Online Sales Data (3).csv

	YearMonth	MonthlyRevenue	OrderVolume	
0	2024-01	14548.32	31	
1	2024-02	10803.37	29	
2	2024-03	12849.24	31	
3	2024-04	12451.69	30	
4	2024-05	8455.49	31	
5	2024-06	7384.55	30	
6	2024-07	6797.08	31	
7	2024-08	7278.11	27	

	YearMonth	MonthlyRevenue	OrderVolume	
0	2024-01	14548.32	31	
2	2024-03	12849.24	31	
3	2024-04	12451.69	30	
1	2024-02	10803.37	29	
4	2024-05	8455.49	31	

	YearMonth	MonthlyRevenue	OrderVolume	
0	2024-01	14548.32	31	
2	2024-03	12849.24	31	
6	2024-07	6797.08	31	
4	2024-05	8455.49	31	
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Next steps:

[Generate code with monthly_stats](#)

[New interactive sheet](#)

[Generate code with top_revenue](#)

[New interactive sheet](#)

[Generate code with top_orders](#)

